

Chapter 10

Grammaticalization of future markers in Korean

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Contemporary Korean has a number of forms that make the future time reference. Their grammatical statuses are variable, e.g., an adnominalizing suffix, modal auxiliary, sentence-ender, etc. Unlike the cross-linguistically attested patterns, Korean futures involve two grammatical formants that inherently involve futurity, i.e., suffixes indicating ‘(at) a later time’ and ‘mode, manner, purpose’, which are primarily responsible for the emergence of future markers. In structural terms, the omissibility of sentential arguments and agglutinating morphology play an important role, because they make the grammatical status of a construction undergoing grammaticalization ambiguous and consequently promotes functional reinterpretation and structural reanalysis. A review of grammaticalization scenarios with respect to the grammaticalization parameters shows that grammaticalized future markers exhibit variable degrees of conformity to them.

1 Introduction

Whether Korean has future tense markers or not remains controversial. Early studies propose a tripartite system of distinguishing the past, the present, and the future (Joo 1910; Park 1972[1935]; Choe 1937; Hong 1947; Yoo 1956; among others), but later studies, notably from Ko (1965) and onward (see also Nam 1972; Sohn 1975; Sung 1976; Suh 1976, 1977; Mun 2005, 2009; among others), tend to subscribe to the binary tense distinction (the past and the non-past), placing the future in the category of modality, based on the fact that the alleged future markers signal such modal notions as conjecture, inference, intention, etc. However, there is still



a body of literature that subscribes to the tripartite tense distinction, e.g., Choi (1995, 1996), Hong (1987, 2008), Min (1991, 1998), among others.

The categorial nomenclature notwithstanding, Korean does have a number of markers that signal the ‘futurity’ of an event or state of affairs, hence ‘future markers’ from a broad perspective. These markers exhibit different historical depths, degrees of grammaticalization, levels of currency, and areas of specialization. This chapter addresses the development of those markers from a grammaticalization perspective.

Korean is spoken by 81.7 million people (Eberhard et al. 2023) mostly in and around the Korean peninsula. It was formerly regarded as an Altaic language as the eminent Finnish Altaic linguist Gustaf Ramstedt proposed in 1939 (Ramstedt 1997[1939]), but in the modern linguistic research tradition, the Altaic hypothesis is contested (see Kim 1992; Sohn 1999; Song 2005, among others, for discussion of the issue). *Ethnologue* (26th ed., Eberhard et al. 2023) lists Korean as a Koreanic language along with Jejueo, a language variety spoken in Jeju Island. Recently, it is frequently discussed under the geographically motivated classificatory term ‘Transeurasian languages’ (Johanson & Robbeets 2010), often along with Japanese, since the two languages share many structural characteristics (Narrog & Rhee 2013; Narrog et al. 2018; Robbeets et al. 2021; see Tian et al. 2022 and Robbeets et al. 2022 for controversy).

The historical depth of texts written in *Hangeul* (or *Hankul*), the Korean alphabet, goes back to the 15th century CE (1443). The invention of this alphabetic-featural script is a landmark event that divides Early Middle Korean (EMiK; the 10th c. through the mid-15th c.) and Late Middle Korean (LMiK; the mid-15th c. through the 16th c.), and a large body of texts has been electronically archived through government-led projects. Prior to the invention of *Hangeul*, several different writing systems were invented using Chinese characters for their meaning (semantogram) or sound (phonogram) to represent Korean words or grammatical morphemes.

Typologically, Korean is an agglutinating, head-final, SOV language. In line with the SOV word order characteristics in morphology, it is suffixal and postpositional. Multiple suffixes and postpositions may occupy slots in a relatively fixed order. Argument NPs are often omitted and NPs may occur without their postpositional particles, especially in speech, and interpretive flexibility thus created often affects or effects grammaticalization. In terms of verbal morphology, finite verbs are marked for tense, aspect, mood, and modality, whereas non-finite verbs are often marked with ‘converb’-markers (*converbialia*, Ramstedt 1903), the suffixes whose main function is adverbializing a non-finite verb (Haspelmath 1995). Finite verbs occurring sentence-finally are further marked by a sentence-

type marker, such as declarative, imperative, interrogative, hortative, exclamative, etc., all of which forming a complex paradigm further modulated by the degree of honorification and politeness, commonly known as *hwakyey* ‘speech level’ (Rhee 2020).

This chapter is organized as follows: Section 2 addresses the development in form and function of the future markers; Section 3 discusses morphosyntactic and conceptual motivations and reviews the grammaticalization scenarios in view of grammaticalization parameters; and Section 4 summarizes the findings and concludes the chapter.

2 Future time reference in Korean

In LMiK, the notion of future was carried by several different morphosyntactic devices, such as monosyllabic *-li-* and the periphrastic forms involving the mode converb-marker *-key* (e.g., *-key taoy-*, *-key ha-*, and *-kuys-*). However, these markers did not develop their tense-aspectual scopes, and they either survived in limited usages or completely disappeared, in Present-day Korean (PDK; the 21st c.).

PDK has several future (and future-related) markers. Historically, they belong to three developmental trajectories, which form layers of different historical depths, i.e., *-li-*, *-l.ke(s).i-*, and *-keyss-*. Among these, the oldest form is *-li-*, attested in Old Korean (OK; 1st c. through early-10th c.). On the other hand, the other two are more recent innovations; *-l.ke(s).i-* developed from the 16th century (Lee 1997: 64) and *-keyss-* developed in the early 19th century (Huh 1987; Rhee 1996). We discuss those three classes of future markers focusing on their historical development in form and function, in the following.

2.1 The *-li*-Class futures

2.1.1 The future adnominalizer *-l*

The first future marker is *-l*, a future adnominalizer, known as the ‘prospective adnominalizer’ (PROS.ADN) (Rhee 2012b), of which the major function is to transform a verbal or clausal constituent into a modifier of a noun phrase, hence the label ‘adnominalizer’. This function is illustrated in the following:

- (1) a. 올 사람
 o-l *salam*
 come-FUT person
 ‘a person who will come’

- b. 살 옷
sa-l *os*
 buy-FUT clothes
 ‘clothes that (I) will buy’

Examples in (1) shows that *-l* heads a constituent, which modifies the following noun (note that Korean is head-final), e.g. *salam* ‘person’ in (1a) and *os* ‘clothes’ in (1b). Sometimes, structurally complex constituents involving the future *-l* becomes fully unverbated, e.g., *mek-ul kes* [eat-FUT thing] ‘food’ (< ‘thing to eat’); *ip-ul kes* [wear-FUT thing] ‘clothing’ (note that *-l* has a phonologically controlled allomorph *-ul*, which occurs after a coda). As is obvious from the examples above, *-l* is not a finite-verb morphology; it is used regardless of the speech time, as shown in (2), in which *-l* occurs in the sentences marked by the past tense.

- (2) a. 올 사람은 나 왔다.
o-l salam-un ta w-ass-ta
 come-FUT person-TOP all come-PST-DEC
 ‘Everyone who would come came.’
 (PDK, constructed)
- b. 그것은 내가 살 옷이었다.
kukes-un nay-ka sa-l os-i-ess-ta
 it-TOP I-NOM buy-FUT clothes-BE-PST-DEC
 ‘It was the clothes that I would buy.’
 (PDK, constructed)

The examples in (2), in which *-l* occurs in the past-marked sentences, clearly shows that the future *-l* is not a regular future-tense marker, which, by definition, is based on the speech time.¹ Thus, *-l* is a prospective marker (as ‘prospective adnominalizer’, Rhee 2008), one belonging to the ‘relative tenses’ (McCawley 1971: 111; Partee 1973: 609; Comrie 1985; Dahl 1985; Michaelis 2006: 220; Hengeveld 2011), ‘secondary tenses’ (Diewald 2011; Lyons 1977: 689), ‘relational tenses’ (Hopper & Traugott 2003: 152), and ‘anaphoric tenses’ (Lo Cascio 1986: 198), in contrast with the absolute tenses anchored in the moment of speaking. Relative tenses are also called ‘compound tenses’ because these markers are, in general, morphosyntactically and/or semantically complex, by referring back to times denoted by previous temporal expressions (e.g., English *will have done*, *had been doing*, etc.).

¹For this reason, Rhee (2008) describes *-l* 'futuristic' and exclusively uses the label 'prospective adnominalizer', avoiding 'future'.

In this regard, the Korean future *-l* is qualitatively different in that it does not accompany a reference time expression. The future *-l*, by virtue of marking a later point in time, may also be conceived as a ‘posterior’ tense (Jespersen 1924: 257; Bull 1963: 14; Kamp & Rohrer 1983: 254–257), as in the English posterior future ‘will be about to sing’ (Binnick 2001: 559).²

Historically, the future-marking *-l* had functioned as a nominalizer (e.g., gerundivizer, historically *-lh*, with a glottal stop) with futuristic meaning (‘prospective nominalizer’), before it acquired the adnominalizer function (Kim 1957; Lee 1972; Lee 1997; among others). This function survived as late as 1577 (Rhee 2008: 240–242) but has since become completely defunct; its sole function is a prospective adnominalizer (modifier) in Modern Korean. Its formal vestiges are found in certain sentence-enders, e.g., *-lkka* (futuristic question), *-lla* (apprehensional), *-lsey* (apprehensional), in which the futuristic morpheme *-l* is completely fused (cf. for discussion of interrogative markers involving *-l* in OK and EMiK, see Ko 2011: 88–90, 101–127; for discussion on *-l* in apprehensionals, see Yae et al. 2023; Rhee & Kuteva, in press). It has been noted that, crosslinguistically, the secondary grammaticalization from a nominalizer to a modifier (such as relativizer, complementizer, etc.) is fairly common (Yap et al. 2010: 77; Yap & Wang 2011: 66).

2.1.2 The future *-li-*

The source of the future marker *-li-* involves *-l* (prospective nominalizer/ adnominalizer) addressed above. When *-l* was combined with the defective noun *i*, and the copula *i-*, a new portmanteau form *-li-* emerged (Lee 1997: 56–59; Park 1998: 163). As *-l* had the prospective function, *-li-* naturally inherited this function of marking futurity. Its use is exemplified in the following excerpts:³

- (3) a. 天心을 이루오리라.
CHENSIM-ul ilwu-o-li-la
 providence-ACC accomplish-HON-FUT-DEC
 ‘I will fulfill the Providence (Heavens’ will).’
 (LMiK, 1445 *Yongpiechenka* 108)

²The morpheme *-l* is multifunctional, and an anonymous reviewer directs the author’s attention to Lim (2009), who lists the irrealis-marking usage among its diverse functions.

³Some historical data in Korean has mixed script of Chinese characters and *hangeul*, and a general practice in romanization of such texts is to use upper-case for Chinese characters following the modern Korean pronunciation. Since old Korean alphabets are typographically incompatible, Korean scripts are not presented if the excerpts contain such characters.

- b. 내일이면 고향에 다다르리라.
nayil-i-myen kohyang-ey tatalu-li-la
 tomorrow-BE-COND hometown-AT reach-FUT-DEC
 ‘I will arrive at (my) hometown by tomorrow.’
 (MoK, *Wulimalsaym* ‘li’)
- c. 나는 가리라, 곧 가리라 이니스프리섬으로.
na-nun ka-li-la, kot ka-li-la inisuphuli.sem-ulo
 I-TOP go-FUT-DEC soon go-FUT-DEC Innisfree.island-DIR
 ‘lit. I will go, To Innisfree Isle (I) will soon go.’
 ‘org. I will arise and go now, and go to Innisfree.’
 (MoK translation, William B. Yeats, ‘The Lake Isle of Innisfree’)

Examples in (3) show that *-li-* is functioning as a future marker. In each example *-li-* is followed by other verbal morphologies, e.g., a sentence ender, and, thus, the future marker *-li-* is one that occupies a slot known as the penult ender position. Note that the clause containing the second occurrence of *-li-* in (3c) is in the inverted word order for poetic effect. Even though *-li-* was the primary future marker prior to MoK, it is now archaic or poetic.

In terms of functions, the future *-l-* signals intention or determination, as in (3a) and (3c) or conjecture as in (3b). However, as functional classification largely depends on the variable granularity, other functions related to these overarching notions such as prediction, possibility, ability, obligation, etc. have been also marked by *-li-*.⁴

2.1.3 The future sentence-ender *-li*

The next future marker is the sentence-ender *-li*, nearly identical in form and function with the previously addressed *-li-* above. Its use is exemplified in the following:

- (4) a. Declarative
YENG CWU-z alph-Ay naynay pwuskuli-li
 wise.king-GEN front-AT forever be.ashamed-FUT.DEC
 ‘(The unwise king) will be ashamed forever in the face of a wise king.’
 (LMiK, 1447 *Yongpiechenka* 16)

⁴Lee (1997: 32–35) also reports that in LMiK, the future *-li-*, in combination with a negative adverb *mot* ‘not’ or a negative verb *mal-* ‘stop; do not’, marked prohibition, premonition, etc. (see Lee 1997: 35).

b. Interrogative

CENGSAISEONG-skuy nw-i ani o-ZAVA-li

national.hero-DAT who-NOM not come-HON-FUT.Q

‘Who would not follow this hero who defended the nation?’

(LMiK, 1447 *Yongpiechenka* 99)

This future marker *-li* is different from the previously addressed *-li-* in that it is the sentence-ender (i.e., ultimate verbal morphology), whereas the latter is the one occurring in the penult verbal morphology. This is a unique situation because tense markers occur in the penult position and the ultimate position is reserved for sentence-type markers. The sentence-ending future markers occur only in declarative and rhetorical interrogative sentence types, as exemplified in (4) above. Since it is directly derived from the previous *-li-*, the future sentence-ender *-li* has the same functions, i.e., notably conjecture and intention, as well as other future-related functions.

This marker occurs very infrequently in the historical data, i.e., only in three sources that are written in the poetic style, i.e., *Yongpiechenka* (1447), *Welinchenkangcikok* (1447), and *Welinsekpo* (1447) (Yeom 1998: 20). In MoK, the use of this future sentence-ender *-li* is still restricted to poetic genres. For this reason, this future-marking sentence ender has not been discussed in earnest (Ko 1990 and Yeom 1998 are notable exceptions). The development of this unique marker results from a rather simple process. Sentence-final particles formerly attached to the tense-marking *-li-* were omitted, presumably for poetic effects (e.g., to match the number of syllables with the counterpart stanza, cf. Yeom 1998: 20–21), and, consequently, the suffix was formally reanalyzed and functionally reinterpreted as a future-marking sentence-ender. For the lack of verbal morphologies that normally follow the tense marker, Ko (1990: 265) and Ryu (1992: 422) regard this ender as an instance of the *panmal* ‘half-talk’ style, i.e., one without honorification or politeness markers.

2.2 The future *-l.ke(s).i-*

The second group of future markers consists of two historically related forms, *-l.kes.i-* and *-l.key*; the latter has diverged from the former. The future *-l.kes.i-* developed from around the 16th century (Lee 1997: 64) and is productively used in MoK.

2.2.1 The future *-l.kes.i-*

The future tense marker *-l.kes.i-* is a periphrasis involving the prospective (i.e., future-marking) adnominalizer *-l* (see Section 2.1.1 above), the formal noun (i.e., semantically vacuous) *kes* ‘thing’, and the copula *i-* ‘be’.⁵ This complex construction has become fully univerbated through structural reanalysis and functional reinterpretation (see Section 3.1 below for more discussion). The future *-l.kes.i-* is exemplified in the following:

- (5) a. 버스가 곧 올 것이다.
pesu-ka kot o-l.kes.i-ta
 bus-NOM soon come-FUT-DEC
 ‘The bus will come soon.’

(Lee & Lee 2010: 500)

A brief note on the formal noun *kes* is in order. *Kes* is among the most frequently used nominalizers in MoK. It started its life as a full-fledged noun denoting ‘thing, skin, surface’ (Hong 1983; Rhee 2011) but became a defective noun from LMiK (Huh 1983, Jeong 1989; Lee 1988), labeled as such because it is so semantically weak that it cannot be used independently (also called ‘formal noun’). Even though it is semantically bleached, syntactically it still functions as the head noun of a nominalized string (Rhee 2011: 395), and thus requires that the modifying constituent be headed by an adnominalizer (such as the prospective adnominalizer *-l* here).

As the complex periphrasis *-l.kes.i-* becomes univerbated into a single grammatical unit of signaling future, its function also develops into the conceptual domain marked by its earlier, and more strongly entrenched, competitor *-li-*. Thus, *-l.kes.i-* marks conjecture, intention, determination, possibility, prediction, obligation, etc. (see Section 2.1.2 above), largely depending on the context (in particular, the sentential subject) and the speech situation, as exemplified below (note that *-ul* is an allomorph of *-l*):

⁵The lexical origin of the noun *kes* is the noun *kach/kas/kes* ‘skin, fur, surface, appearance, thing’ (Hong 1983; Rhee 2008: 248), which have been completely bleached. As an anonymous reviewer points out, the noun *kes* in *-l.kes.i-* may be replaced by other nouns, e.g., *seym* ‘plan’, *sayngkak* ‘thought’, etc., which form predicative constructions known as ‘mermaid’ constructions (Tsunoda 2013; Kim 2013; Rhee 2020: 598–599). Since these nouns still retain their lexical meaning, the mermaid constructions are regarded as syntactic constructions, unlike *-l.kes.i-*, which is a relatively ‘pure’ future marker. The future meaning in the mermaid constructions is derived from the future adnominalizer *-l*, discussed in Section 2.1.1.

- (6) a. prediction, conjecture, conviction

내일도 추을 것이다.

nayil-to chwuw-ul.kes.i-ta

tomorrow-ALSO be.cold-FUT-DEC

‘Tomorrow will be cold, too.’

(Lee & Lee 2010: 500)

- b. intention, determination

나는 꼭 성공할 것이다.

na-nun kkok sengkongha-l.kes.i-ta

I-TOP by.all.means succeed-FUT-DEC

‘I will succeed by all means.’

(Lee & Lee 2010: 501)

In MoK, *kes* [kʌs] itself has undergone phonological erosion and is often used in the form of *ke* [kʌ] (note that *-l.ke(s).i-* is only one of many other lexical and grammatical forms *kes* participates in). Thus, the future marker *-l.kes.i-* has its phonologically reduced variant *-l.ke.i-*, from which the copula *i-* may also be eroded due to the V-V sequence, which triggers deletion of one vowel or modification of the vowel quality. This process results in *-l.ke-*, and thus the verb forms in (6) may also be *chwuw-ul.ke-ta* ‘will be cold’ and *sengkongha-l.ke-ta* ‘will succeed’, respectively. There are no functional differences between these fuller and eroded variants, other than that the reduced variant is more closely associated with the colloquial styles.

2.2.2 The future *-l.key*

The future *-l.key* originates from the reduced form of the future *-l.kes.i-*, described above. Its usage is exemplified below:

- (7) a. promissive

내가 맛있는 것 사 줄게.

nay-ka masissnun kes sa-cwu-l.key

I-NOM delicious thing buy-BEN-PROM

‘I will buy something delicious for you.’

(Lee & Lee 2010: 501)

- b. polite command/request

여기 침대에 누우실게요.

yeki chimtay-ey nwuw-si-l.key-yo

here bed-AT lie.down-HON-IMP-POL

‘Please lie on your back on the bed over here.’

(Koo & Rhee 2013: 489)

As shown in (7), *-l.key* is the sentence ender, i.e., occurring at the ultimate position (note that politeness marker *-yo* is not considered in slot assignment), thus different from *-l.kes.i-* occurring at the penult position, described in the previous section.

When the reduced form is further marked by the ‘informal’ speech level, ‘universal’ speech-act sentence-ender *-e/a*, the form contains a series of vowels as in *-l.ke.i-a* /l.kʌ.i-a/, a situation highly susceptible to vowel change in Korean, thus resulting in *-l.key* /l.ke/. This shows that the fusional process obliterates the word boundary as well as morpheme boundaries. The earliest attestations of *-lkey* (e.g., 17th c. *Pakssipwuincen* movable type ed.) carried the same function as *-l.kes.i-*, i.e., conjecture, prediction, intention, etc. (cf. (Koo & Rhee 2013: 490)), but the reduced form gradually diverged, and from around the turn of the 20th century, it became increasingly specializing as a promissive marker (‘I promise that...’) (e.g., 1911 *Moktanhwa*; cf. Koo & Rhee 2013: 490).

In the case of promissive, as in (7a), since the speaker is also the agent of the proposed act, the promissive sentence is compatible with the first person subject only (either singular or plural). A Korean idiosyncrasy is that the understood subject may freely be omitted as is the case with other sentential constituents. Naturally, in promissive sentences, the implicit subject is the speaker, and the verb cannot be marked with honorification, because self-honorification is not acceptable in any situation. Therefore, when the predicate is marked with honorification, the understood sentential subject cannot be the speaker. When such forms are encountered in a sentence in which a putative promissive marker *-l.key* is used, as in (7b), in a situation where a nurse is giving an instruction to a patient, the addressee will make a pragmatic inference that since it cannot be the promissive sentence, it should be some other speech act. In other words, due to the omissibility of the sentential subject, the only signal that forces the addressee to interpret the sentence as an imperative is the presence of the honorification marker. Thus, the functional reinterpretation of the promissive as the imperative was actualized through pragmatic inference. Through this pragmatic inference, new functions of request (a milder form of imperative) or a command (a stronger form of imperative) emerged in PDK.

2.3 The future *-keyss-*

In PDK there are two primary futurity markers, *-l.kes.i-* and *-keyss-*. As shown in Section 2.2, *-l.kes.i-* etymologically involves the noun *kes* ‘thing’. The latter, *-keyss-*, which developed in the early 19th century (Huh 1987; Rhee 1996), historically involves a number of morphemes (see below), especially, the mode converb-

marker *-key* ‘in such a manner that’, largely denoting mode, manner, purpose, etc. Its uses are illustrated in the following examples taken from Lee & Lee (2010: 67) with modification:

- (8) a. immediate future
 잠시 후에 세 시가 되겠습니다.
camsi hwu-ey sey si-ka toy-keyss-supnita
 moment after-AT 3 hour-NOM become-FUT-DFR.DEC
 ‘It will be 3 o’clock momentarily.’
- b. conjecture
 어서 가자, 학교에 늦겠다.
ese ka-ca, hakkyo-ey nuc-keyss-ta
 fast go-HORT school-TO be.late-FUT-DEC
 ‘Let’s hurry up; we will/might be late for school.’
- c. possibility
 나도 그만큼은 먹겠다.
na-to ku.mankhum-un mek-keyss-ta
 I-ALSO that.much-TOP eat-FUT-DEC
 ‘(I think) I, too, can eat that much. (It is possible that I...)’
- d. intention
 내일은 미국으로 떠나겠어.
nayil-un mikwuk-ulo ttena-keyss-e
 tomorrow-TOP America-DIR leave-FUT-END
 ‘I will leave for America tomorrow.’

Example (8a) marks the immediate future, thus a type of the ‘metrical tense’ (Chung & Timberlake 1985: 207ff; Comrie 1985: 83; Dahl 1985: 120ff; Binnick 2001: 560, among others). Even though the functions of *-l.kes.i-* and *-keyss-* overlap to a great extent, *-keyss-* tends to specialize in immediate future (see Section 3.2 for discussion on its conceptual motivation). Example (8b) makes reference to conjecture as to the future event. Since conjecture can also make reference to the past event, *-keyss-* is not restricted to future events (e.g., ‘They must have been late for school yesterday.’), in which case the future marker *-keyss-* follows the past tense marker *-ess-*. Example (8c) signals possibility but it may also mark its closely related concept ‘ability’, depending on the locus of the enabling force, i.e., within a sentient being (ability) or in the situation (possibility). Example (8d) marks the intention, typically the speaker’s in declarative and the addressee’s in interrogative.

Even though the examples in (8) show the common functions of the future tense *-keyss-*, there are a number of other functions that may be subsumed in those categories at varying degrees, e.g., ability, hypothetical willingness, current state as futuristic mode, evaluative, predestination, among others (Rhee 1996: 109–130).

The development of *-keyss-* (and its dialectal form *-kes(s)-*) has been studied by a number of researchers, from as early as Underwood (1890), Choe (1989 [1929]), Park (1972[1935]), and Ramstedt (1997[1939]) (see also Lee 1947; Na 1953; Shin 1972; Suh 1977; Gim 1981; Rhee 1996; Lee 1997, 2006). The seemingly simple form *-keyss-* originates from a construction involving multiple morphemes, i.e., the mode converb-marker *-key* ‘in such a manner that’, the light verb *ha-* ‘be/do/say/make’, the sequential converb-marker *-e* ‘and’, the existence verb *iss-* ‘exist’, even though many of them are not visible in the current form *-keyss-*. The long process can be briefly described below.

The first is the emergence of the resultative/perfect/past marker *-ess-*, developed from the converb-marker, also known as the ‘consolidating connective’ *-e/a* (Koo 1987), and the verb of existence *-is(s)*, through a long process from the 15th c. The consolidating connective, largely translatable as ‘and’, connects two sequentially related events and makes one holistic event (e.g., *pel-e-mek* ‘earn’ and ‘eat’ > ‘make a living’; *kuw-e-mek* ‘roast’ and ‘eat’ > ‘eat roasted meat’; etc.). By virtue of the presence of the existence verb in its historical origin, *-ess-* tends to imply a strong relevance of a completed event to the present time (Note that it was a resultative marker before it developed into the perfective/past).

The second is the emergence of the periphrastic causative *-key.ha-*, developed from the mode converb-marker *-key* and the light verb *ha-* ‘do/be/say/make’. The converb-marker *-key* is inherently futuristic because it refers to a resultant state of an action or event, e.g., *aphu-key* [ache-MOD] ‘in such a manner that one would feel pain; painfully’, *cwuk-key* [die-MOD] ‘in such a manner that one would die; leading to death’, ‘fatally’ etc. The verb *ha-* is a semantically bleached verb (thus the ‘light’ verb) with various general meanings. The combination of the ‘mode’ and ‘make’, thus ‘make x in such a manner that y will’, led to the development of the causative.

The finally derived sense from prospective causality, definiteness, and current relevance is the standard notion of futurity, in that the speaker is making a judgment with respect to the future, based on the current definite situation which is a result of a completed event. Therefore, the notion ‘conjecture’ is the most basic one in this marker.

3 Discussion

In the preceding section, we have seen the development of various future markers, which can be categorized in three groups, one related to the prospective adnominalizer *-l* (i.e., *-l*, *-li-*, and *-li*), one involving the defective nominal *kes* ‘thing’ (i.e., *-l.kes.i-* and *-l.key*), and one involving the mode converb-marker *-key* (i.e., *-keyss*). Now we turn to a discussion of morphosyntactic motivation (Section 3.1), conceptual motivation (Section 3.2), and grammaticalization parameters (Section 3.3).

3.1 Morphosyntactic motivation

There are a number of prominent factors of the development of the future in terms of morphosyntactic motivation. Among them are having a verb as the last component of the construction, the omissibility of arguments, and agglutinating morphology.

We can see that the copula is involved in the development of most future markers, e.g., *-li-*, *-li*, *-l.kes.i-*, *-l.ke.i-*, and *-l.key*. In these markers, the last element *i-* originates from the copula *i-*, even though it is not directly visible in *-l.key* (see Section 2.2.2). The copula in the source constructions of these markers enables the preceding nominals/nominalizers (i.e., *l* and *kes*) to function as equative or identificational predicates by being a part thereof. For instance, we noted that the future marker *-l.kes.i-* developed from a complex source construction involving the prospective adnominalizer *-l*, the formal noun *kes* ‘thing’, and the copula *i-* ‘be’, which became fully unverbated through structural reanalysis and functional reinterpretation. This is exemplified in the following:

(9) 그가 올 것이다.

a. source structure

ku-ka o-l kes-i-ta
 he-NOM come-PROS.ADN thing-BE-DEC
 ‘lit. It is (a thing) that he comes at a later time.’

b. grammaticalized structure

ku-ka o-l.kes.i-ta
 he-NOM come-FUT-DEC
 ‘He will come.’

The structural reanalysis of (9a) into (9b) involves a number of noteworthy aspects. As is obvious from the translation of (9a), the entity being equated with

‘thing’ in the predicate is not the sentential subject ‘he’ (note that *kes* ‘thing’ is not used to refer to a person in Korean). The sentential subject is not explicit, i.e., the source structure of (9a) is a subject-less sentence, an idiosyncrasy of Korean syntax. The implicit subject is ‘the current situation’, i.e., the current situation is such that ‘he comes at a later time’. What aspects of the current situation warrants the conjecture are not explicit and thus unclear – they are only vaguely referred to as a ‘thing’, a perfect reference as its semantics is nearly vacuous (thus called a formal/defective noun, functioning as a nominalizer). Thus, the absence of concrete semantic content in *kes* ‘thing’, the propinquity between the prospective adnominalizer *-l* and *kes*, the copula following *kes*, and the presence of a contentful verb preceding all these components, among others, created a perfect syntagmatic condition for the reanalysis of the complex construction *-l.kes.i* as an auxiliary (note that the syntactically designated position of an auxiliary is between the main verb and the sentence-type marker like the declarative *-ta* in 9).

It is also noteworthy that the reanalysis illustrated in (9) involves ‘syntactic upgrading’ (Heine et al. 1991: 169–170; Ramm 2008; Rhee 2012a: 297–298), i.e., the formerly subordinating clause (‘he comes at a later time’) becomes the main clause with the future-tense interpretation (‘he will come’). Reanalysis of a former subordinate clause into a main clause typically occurs with ‘insubordination’ (Evans 2007), i.e., main clause ellipsis, which is responsible for many instances of grammaticalization in Korean, but the type of reanalysis with the future marker described here is actualized when the subject is implicit and its predicate is fused with the predicate of the subordinating clause.

Another type of morphosyntactic motivation is with reference to the future *-keyss-*. As alluded to in Section 2.3, it developed from the string, *-key-ha-e-iss-* [mode-make-and-exist], which developed into [CAUS-PERF]. This is illustrated below:

(10) 곧 비가 오게 하여 있어요.

- a. *kot pi-ka o-key-ha-e-iss-e-yo*
soon rain-NOM come-IN.SUCH.A.MANNER-MAKE-AND-EXIST-END-POL
‘It (the situation) is such a manner as to make the rain come soon and exists.’
- b. *kot pi-ka o-key.ha-ess-e-yo*
soon rain-NOM come-CAUS-PERF-END-POL
‘It has made the rain come soon.’

- c. *kot pi-ka o-keyss-e-yo*
soon rain-NOM come-FUT-END-POL
‘Rain will come soon.’

One caveat is that the morphemic breakdown along with glosses and interpretations in the examples is grossly oversimplified, glossing over the complexity of the development; it is presented as such for the sake of convenience, hence the awkwardness of the interpretation. For instance, there is no instance in historical data that is comparable to (10a), simply because the construction did not string together these exponents all at once at one point of time. In other words, there were always differential degrees of semantic cohesion and formal fusion between the components, which also changed over time in the course of grammaticalization (for detailed description of the processes, see Rhee 1996: 109–130 and Lee 1997: 64–82, among others).

Simplicity of data presentation notwithstanding, the syntagmatic condition of *-keyss-* is remarkably similar to that of *-l.kes.i-*, in that the construction immediately follows a verb and is followed by a sentence-ender, a condition perfect for the exponent to be reanalyzed as an auxiliary. The implicit subject, one that takes the serialized verbs *ha-* ‘make’ and *iss-* ‘exist’ as its predicate, is something vague, such as ‘situation’, and does not surface in the sentence. As the two verbs became invisible due to formal fusion, the verb of the subordinating clause was reanalyzed as the main verb, and accordingly the fused exponent was reanalyzed as the auxiliary, once again an instance of syntactic upgrading.

From the exposition above, it is evident that the syntactic idiosyncrasy in Korean that sentential constituents (the subject in the present case) can be freely omitted has played an important role in the grammaticalization of the future markers. In structural terms, the fact that the last element of the construction is a verb (e.g., *i-* ‘be’ and *iss-* ‘exist’ above) is important because the verb occurs sentence-finally, marked by the sentence-type marker (also known as the sentence-ender). The slot immediately before the sentence-type marker is for TAM, including the future auxiliary. Also significant is the typological characteristic of agglutinating morphology because agglutination and fusion obliterate the morphemic boundary, make the exponent’s grammatical status ambiguous, and consequently promotes functional reinterpretation and structural reanalysis.

3.2 Conceptual motivation

In crosslinguistic studies, the future is known to develop from diverse conceptual sources. For instance, the lexicon by Kuteva et al. (2019) lists Change-of-state,

‘come to’, ‘go to’, ‘love’, B-Necessity, D-Necessity, H-Possessive, ‘take’, ‘then’, ‘tomorrow’, Venitive, and ‘want’, as the commonly attested sources of future markers. In their seminal work, Bybee et al. (1994: 251–280) show that lexemes related to desire (‘want’, ‘like’), ability (‘be able to’), obligation (‘owe’), movement verbs (‘come’, ‘go’), temporal adverbs (‘then’, ‘thereafter’, ‘afterward’, ‘just now’, ‘soon’), possession (‘have’, ‘get’, ‘obtain’, ‘catch’) as well as other semantically general verbs (e.g., ‘be’, ‘become’, ‘try to’, ‘need’, ‘owe’, ‘get’, ‘do’, ‘make’, ‘look for’, ‘intend’), etc. develop into the future across languages.

The Korean paths, however, are difficult to reconcile with these known cross-linguistic patterns. Even though the Korean verb *ha-* ‘make, do’ may be close to the general verbs in the inventory, and the copula *i-* ‘be’ is listed in it as well, we noted in the above that they do not make a notable contribution to the development. They are similar to syntactic placeholders rather than semantic contributors. Far more important contributors are the prospective adnominalizer *-l* (e.g., ‘book to read’), which is already inherently futuristic by virtue of indexing a later point in time, and the mode converb-marker *-key* (e.g., ‘painfully’), which inherently signals the effect- or goal-orientation (Rhee 1996). As briefly alluded to in Section 2.3, the converb suffix *-key* marks ‘manner’, ‘mode’ or ‘purpose’. The manner/mode of a situation or the purpose of an action has direct implication to the future because they are the bridge between one situation and its future situation or between one action and its future resultant situation. For this reason, the future signaled by *-keyss-*, as compared to *-l.kes.i-* or *-li-*, is close to the present time, hence, the immediate future. Even though the relation between manner/purpose and future is almost universal, such relation implied by *-key* in Korean is so prominent that (Ramstedt 1997[1939]: 91) calls the verbs connected by the morpheme *-key* ‘converbium futuri’, a position tantamount to classifying *-key* as a full-fledged future marker.

All the conceptual contributors to the development of Korean future markers can be shown as a diagram in Figure 1, which is necessarily brief.

3.3 Grammaticalization parameters

The grammaticalization scenarios of Korean future markers can be reviewed with respect to the four grammaticalization parameters suggested in Kuteva et al. (2019), i.e., desemanticization, extension, decategorialization, and erosion.

3.3.1 Desemanticization

Desemanticization refers to the loss of meaning content. Even though it is difficult to compare the source (or earlier) and target meanings, a rough generalization from the foregoing exposition can be presented as in Table 1.

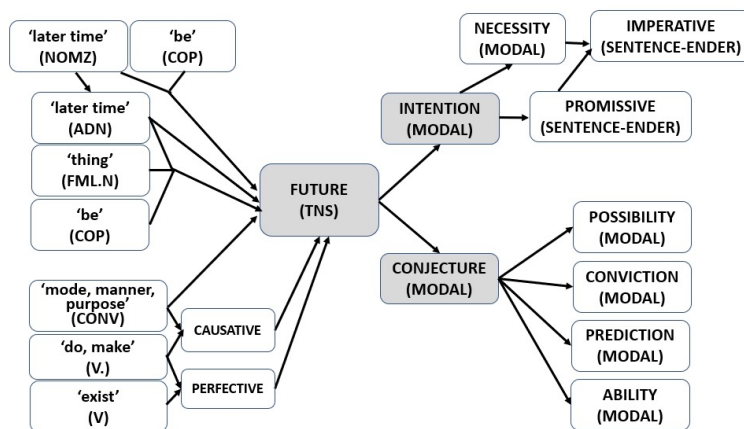


Figure 1: Conceptual contributors to Korean future markers

Table 1: Desemanticization in grammaticalization of future markers

Form	Source meaning (literal)	Target meaning	Desemanticization
-l	'later time'	Future	Not prominent
-li-	'be (at) a later time'	Future	Not prominent
-li	'be (at) a later time'	Future	Not prominent
-l.ke(s).i-	'it is the thing that P at a later time'	Future	'thing' disappeared
-l.key	'it is the thing that P at a later time'	'I promise that...' 'Please do x'	'thing' disappeared
-keyss-	'X is in the mode of Y and exists'	Future	'mode' bleached, 'exist' bleached

Table 1 shows that desemanticization in the *-li*- Class futures is not prominent, because they started out (at least from the earliest available data) as a word/morpheme denoting a later time (either as a nominalizer or adnominalizer). The words denoting ‘thing’ (*kes*) and ‘exist’ (*iss-*) as well as the converb-marker with the ‘mode’ meaning (*-key*) have their source meanings either completely absent or partially bleached.

3.3.2 Extension

Extension as a grammaticalization parameter refers to the use of a form in new contexts. A comparison of the use contexts of the source structure to those of the current, future usage can be presented as Table 2.

Table 2: Extension in grammaticalization of future markers

Form	Source context	Target context	Extension
<i>-l</i>	Verbal morphology (Nominalizer)	Verbal morphology forming a pre-nominal modifier	No extension (context shift)
<i>-li-</i>	Verbal morphology for predicate-formation (cf. <i>-l-i-</i>)	Verbal morphology, modal suffix	No extension (context shift)
<i>-li</i>	Verbal morphology, modal suffix	Verbal morphology, sentence ender	No extension (context shift)
<i>-l.ke(s).i-</i>	Verbal morphology involving ‘a thing’	Verbal morphology without referential restriction	Extension
<i>-l.key</i>	Verbal morphology involving ‘a thing’	Verbal morphology without referential restriction	Extension
<i>-keyss-</i>	Verbal morphology involving causative and perfective meaning	Verbal morphology with no causative/perfective meaning	Extension/Shift

As shown in Table 2, the future markers involving the lexeme *kes* ‘thing’ (i.e., *-l.ke(s).i-* and *-l.key*) have undergone extension in that the future does not have to involve reference to an entity. Similarly, *-keyss-*, which contains elements denoting causative and perfective in earlier stages, can be used without such restrictive meanings. Extension, by definition, means an increase in use contexts and, thus, implies the availability of earlier usage at a later stage. However, the future markers involving *-l* (i.e., *-l*, *-li-*, and *-li*) do not retain earlier usage, and thus exhibit usage context shift, rather than extension.

3.3.3 Decategorialization

Decategorialization refers to the loss in morphosyntactic properties characteristic of lexical or other less grammaticalized forms. Korean future markers display decategorialization at varying degrees, as shown in Table 3.

Table 3: Decategorialization in grammaticalization of future markers

Form	Source category (feature)	Target category (feature)	Decategorialization
<i>-l</i>	Nominalizer, verbal suffix	Adnominalizer, verbal suffix	Decategorialization Loss of argument function
<i>-li-</i>	(periphrastic) (cf. <i>-l-i-</i>)	Modal auxiliary	Not determinable
<i>-li</i>	Modal auxiliary	Modal suffix (sentence ender)	Decategorialization Loss of ability to be followed by a sentence-ender
<i>-l.ke(s).i-</i>	(periphrastic)	Modal auxiliary	Not determinable
<i>-l.key</i>	(periphrastic)	Modal suffix (sentence ender)	Not determinable
<i>-keyss-</i>	(periphrastic)	Modal auxiliary	Not determinable

The presence/absence or degrees of decategorialization should be based on a single, major (or non-minor) category lexeme. However, as shown in Table 3, most future markers are periphrastic at the source stage, and, thus, decategorialization is largely indeterminable with these forms. Though trivial in extent, some level of decategorialization is observed with *-l* and *-li*. As we discussed in Section 2.1.1, *-l* was a nominalizer, which later became an adnominalizer. As a nominalizer it can form a nominal constituent, which can function as a verbal argument. As it becomes an adnominalizer (i.e., modifier), such a nominal function is lost, an instance of decategorialization. The sentence-ender *-li*, occupying the ultimate slot, is a development from the modal auxiliary *-li-*, occupying the penult slot. Since modal auxiliaries are suffixes that are further marked by sentence-enders (i.e., sentence-type markers), by becoming a sentence-ender, *-li* has lost such earlier categorial function.

3.3.4 Erosion

Erosion refers to the reduction of phonological volume of a grammaticalizing form. Korean future markers are eroded at varying degrees, as shown in Table 4.

Table 4: Erosion in grammaticalization of future markers

Form	Source form	Target form	Erosion
<i>-l</i>	/l/	/l/	No erosion
<i>-li-</i>	/li/	/li/	No erosion
<i>-li</i>	/li/	/li/	No erosion
<i>-l.ke(s).i-</i>	/l.kʌs.i/	/l.kʌs.i/ /l.kʌ.i/	No erosion Eroded (moderate)
<i>-l.key</i>	/l.kʌs.i.a/	/l.ke/	Eroded (extensive)
<i>-keyss-</i>	/ke.ha.ʌ.is/	/kes/ /kʌs/ (dialectal)	Eroded (extensive) Eroded (extensive)

It should be noted that there is a discrepancy between the character-based romanization and phonemic representation of the markers. From Table 4, it is evident that the erosion is variable. The eroded forms are only those that involve the noun *kes* ‘thing’ /kʌs/ and the mode-marking converb connective *-key* /ke/.

From the foregoing exposition, it is evident that the grammaticalization scenarios of Korean future markers exhibit differing degrees of conformity to the grammaticalization parameters, proposed by Kuteva et al. (2019). Some of them are indeterminable because of multiple forms participating in the source construction.

4 Summary and conclusion

This chapter addressed the grammaticalization of Korean future markers. Contemporary Korean has six forms that make the future time reference, i.e., *-l*, *-li-*, *-li*, *-l.ke(s).i-*, *-l.key*, and *-keyss-*. Their grammatical statuses are variable from an adnominalizing suffix (*-l*), to a modal auxiliary (*-li-*, *-l.ke(s).i-*, *-keyss-*), and to sentence-ender (*-li*, *-l.key*).

Unlike the crosslinguistically attested patterns of grammaticalization sources, e.g., lexemes denoting desire, ability, obligation, movement, time, possession, among others, the development of Korean futures is largely due to two grammatical formants that inherently involve futurity, i.e., suffixes indicating ‘(at) a later time’ (*-l*) and ‘mode, manner, purpose’ (*-key*).

In structural terms, a number of factors contribute to the development of future markers, but, most notably, omissibility of sentential arguments and agglutinating morphology play an important role, because they make the grammatical status of a construction undergoing grammaticalization ambiguous (i.e., through omission and fusion), and consequently promotes functional reinterpretation and structural reanalysis.

A review of grammaticalization scenarios with respect to the grammaticalization parameters, i.e., desemanticization, extension, decategorialization, and erosion, shows that future markers exhibit variable degrees of conformity to them in the course of their grammaticalization.

Abbreviations

ACC	accusative	IMP	imperative
ADN	adnominalizer	LMIK	Late Middle Korean
BEN	benefactive	MOD	Mode-converb
CAU	causative	MOK	Modern Korean
COND	conditional	N	noun
CONV	converb-marker	NOM	nominative
COP	copula	NOMZ	nominalizer
DAT	dative	PDK	Present-Day Korean
DEC	declarative	PERF	perfective
DFR	deferential	POL	polite
DIR	directional	PROM	promissive
END	sentence-ender	PROS	prospective
FML	formal	PST	past
FUT	future	Q	question/interrogative
GEN	genitive	TNS	tense
HON	honorific	TOP	topic
HORT	hortative	V	verb

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References

- Binnick, Robert I. 2001. Temporality and aspectuality. In Martin Haspelmath, Ekkehard König, Wulf Oesterreicher & Wolfgang Raible (eds.), *Language typology and language universals*, 557–567. Berlin: Walter de Gruyter.
- Bull, William E. 1963. *Time, tense, and the verb*. Berkeley: University of California Press.
- Bybee, Joan L., Revere Perkins & William Pagliuca. 1994. *The evolution of grammar: Tense, aspect and modality in the languages of the world*. Chicago: Chicago University Press.
- Choe, Hyon-Bae. 1937. *Wuli malpon* [A Korean grammar]. Seoul: Yeonhee University Press (Yonsei). (In Korean).
- Choe, Hyon-Bae. 1989 [1929]. *Wuli malpon* [A Korean grammar]. Seoul: Jungum Publishing. (In Korean).
- Choi, Dong Ju. 1995. *A study on the historical change of the tense-aspect-mood system in Korean*. Korea: Seoul National University. (Doctoral dissertation). (In Korean).
- Choi, Dong Ju. 1996. The historical change of the pre-final ending *-ta* in Korean. *Enehak* 19. 375–409. (In Korean).
- Chung, Sandra & Alan Timberlake. 1985. Tense, aspect and mood. In Timothy Shopen (ed.), *Language typology and syntactic description*, vol. 3, 202–258. Cambridge: Cambridge University Press.
- Comrie, Bernard. 1985. *Tense* (Cambridge Textbooks in Linguistics). Cambridge: Cambridge University Press.
- Dahl, Östen. 1985. *Tense and aspect systems*. Oxford: Basil Blackwell.
- Diewald, Gabriele. 2011. Grammaticalization and pragmaticalization. In Heiko Narrog & Bernd Heine (eds.), *The Oxford handbook of grammaticalization*, 450–461. Oxford: Oxford University Press.
- Eberhard, David M., Gary F. Simons & Charles D. Fennig (eds.). 2023. *Ethnologue: Languages of the world*. 26th. Dallas, Texas: SIL International.
- Evans, Nicholas. 2007. Insubordination and its uses. In Irina Nikolaeva (ed.), *Finiteness: Theoretical and empirical foundations*, 366–431. Oxford: Oxford University Press.
- Gim, Cha-gyun. 1981. Kwankyeyceluy siceywa sangwimwun sokeyseuy yensan 1 [the tense in relative clauses and the operation in the main sentence]. *Han-Geul* 207. 21–60. Reprinted in Gim, Cha-gyun 1993 [1990], *Wulimal siceywa sanguy yenkwu* [A study on Korean tense and aspect]. Seoul: Thayhaksa. (In Korean).

- Haspelmath, Martin. 1995. The converb as a cross-linguistically valid category. In Martin Haspelmath & Ekkehard König (eds.), *Converbs in cross-linguistic perspective. Structure and meaning of adverbial verb forms: adverbial participles, gerunds*, 1–55. Berlin: de Gruyter.
- Heine, Bernd, Ulrike Claudi & Friederike Hünemeyer. 1991. *Grammaticalization. A conceptual framework*. Chicago: Chicago University Press.
- Hengeveld, Kees. 2011. The grammaticalization of tense and aspect. In Heiko Narrog & Bernd Heine (eds.), *The Oxford handbook of grammaticalization*, 580–594. Oxford: Oxford University Press.
- Hong, Jongseon. 1983. Myengsahwa emiuy pyenchen [historical change of nominalizing particles]. *Korean Language and Literature* 89. 31–89.
- Hong, Jongseon. 1987. Kwuke siceyuy paltal [the development of tenses in Korean], special issue, *Sekhen Cengkyupok paksa hwalyekkinymchonglon* [Festschrift for the 60th birthday of sekhen chung kyu bok]. *Emwunnoncip* (Mincokemwunhakhoy) 27. 805–822. (In Korean).
- Hong, Jongseon. 2008. The system of tense morphemes and their function changing in Korean. *Han-Geul* 282. 97–123.
- Hong, Ki-Moon. 1947. *Cosenmwunpep yenkwu* [A study of Korean grammar]. Seoul: Seoul Daily News Publishing. (In Korean).
- Hopper, Paul J. & Elizabeth C. Traugott. 2003. *Grammaticalization*. 2nd edn. Cambridge: Cambridge University Press.
- Huh, Wung. 1983. *15-seyki wuli yeysmalpon* [A 15th century Korean grammar]. Seoul: Saem Publishing. (In Korean).
- Huh, Wung. 1987. *Kwuke ttaymaykimpepuy pyenchensa* [A developmental history of Korean tense system]. Seoul: Saem Publishing. (In Korean).
- Jeong, Howan. 1989. *Hwuki cwungseykwuke uyconmyengsa yenkwu* [A study on defective nouns in Late Middle Korean]. Seoul: Hakmunsa Publishing. (In Korean).
- Jespersen, Otto. 1924. *The philosophy of grammar*. London: Allen & Unwin.
- Johanson, Lars & Martine Robbeets. 2010. Introduction. In Lars Johanson & Martine Robbeets (eds.), *Transeurasian verbal morphology in a comparative perspective: Genealogy, contact, chance*, 1–5. Wiesbaden: Harrassowitz.
- Joo, Si-Kyung. 1910. *Kwukemwunpep* [A Korean grammar]. Seoul: Bakmunsa Publishing. (In Korean).
- Kamp, Hans & Christian Rohrer. 1983. Tense in texts. In Rainier Bauerle, Christoph Schwarze & Arnim von Stechow (eds.), *Meaning, use, and interpretation of language*, 150–169. Berlin: Gruyter.

- Kim, Jounghmin. 2013. Mermaid construction in Korean. In Tasaku Tsunota (ed.), *Adnominal clauses and the 'mermaid construction': Grammaticalization of nouns*, 249–296. Tokyo: NINJAL.
- Kim, Nam-Kil. 1992. Korean. In William Bright (ed.), *International encyclopedia of linguistics*, 282–286. Oxford: Oxford University Press.
- Kim, Wan Jin. 1957. *-N, -l tongmyengsaury thongsaloncek kinungkwa paltaley tayhaye* [On the syntactic functions and development of the *-n* and *-l* gerunds]. *Kwukeyenkwu* 2. 43–72. (In Korean).
- Ko, Eun-Sook. 2011. *Kwukeye uymwunpep emiuy yeksacek pyenchen* [A historical development of Korean interrogative suffixes in Korean]. Seoul: Hankook Publisher. (In Korean).
- Ko, Yong-Kun. 1965. Hyuntaykwukeuy sepepcheikyeyey tayhan yenkwu [a study of modality in modern Korean]. *Kwukeyenkwu* 15. Reprinted in Ko, Yong-Kun, 1989, *Kwukeye hyengthaylon yenkwu* [A study of Korean morphology], 143–224. (In Korean).
- Ko, Yong-Kun. 1990. *Phyocwun cwungseykwuke mwunpeplon* [A grammar of Middle Korean]. Seoul: Tower Press. (In Korean).
- Koo, Hyun Jung. 1987. Ssikkuth ‘-a, -key, -ci, -ko’-uy ssuimkwa uymi [the usage and semantics of suffixes, *-a*, *-key*, *-ci*, and *-ko*]. *Konkuk Emwunhak* 11-12. 167–188. (In Korean).
- Koo, Hyun Jung & Seongha Rhee. 2013. ‘i will do it... But i’m asking you to do it’: On the emergence of polite imperative from promissive. *Procedia - Social and Behavioral Sciences* 97. 487–494. DOI: 10.1016/j.sbspro.2013.10.263.
- Kuteva, Tania, Bernd Heine, Bo Hong, Haiping Long, Heiko Narrog & Seongha Rhee. 2019. *World lexicon of grammaticalization*. 2nd edn. Cambridge: Cambridge University Press.
- Lee, Byeong-gi. 1997. *A diachronic study on the future tense forms*. Seoul National University. (MA thesis). (In Korean).
- Lee, Byeong-gi. 2006. *A historical study on the expressions of futurities in Korean*. Seoul National University. (Doctoral dissertation). (In Korean).
- Lee, Ju-Haeng. 1988. *Hankwukeye uyconmyengsaury thongsicek yenkwu* [A diachronic study on Korean defective nouns]. Seoul: Hansaem Publishing. (In Korean).
- Lee, Hee Seung. 1947. *Kwukeyhak nonko* [papers in Korean linguistics]. Seoul: Eulyoo Publishing. (In Korean).
- Lee, Heeja & Jong Hee Lee. 2010. *Emi cosa sacen* [Dictionary of particles]. Seoul: Hankook Publisher. (In Korean).
- Lee, Soong-Nyeong. 1972. *Kwukeyhak yenkwu* [a study of Korean linguistics]. Seoul: Hyungsol Publishing.

- Lim, Dong-hoon. 2009. The grammatical category of Korean adnominal ending *-eul*. *Korean Linguistics* 44. 55–81. (In Korean).
- Lo Cascio, Vincenzo. 1986. Temporal deixis and anaphor in sentence and text: Finding a reference time. In Vincenzo Lo Cascio & Co Vet (eds.), *Temporal structure in sentence and discourse*, 191–228. Dordrecht: Foris Publishing.
- Lyons, John. 1977. *Semantics*. Cambridge: Cambridge University Press.
- McCawley, James D. 1971. Tense and time reference in English. In Charles Fillmore & D. Terence Langendoen (eds.), *Studies in linguistic semantics*, 97–113. New York: Holt, Rinehart & Winston.
- Michaelis, Laura A. 2006. Tense in English. In Bas Aarts & April McMahon (eds.), *The handbook of English linguistics*, 220–243. Oxford: Blackwell.
- Min, Hyunsik. 1991. *Kwukeyu sisangkwa sikanpwusa* [Korean tense-aspect and time adverbials]. Seoul: Gaemunsa Publishing. (In Korean).
- Min, Hyunsik. 1998. Sikaneuy nathmalpath [the semantic field of time expressions]. *Han-Geul* 240. 323–354. (In Korean).
- Mun, Suk Yeong. 2005. *Hankwuke sicey pemcwu yenkwu* [A study of the Korean tense category]. Seoul National University. (Doctoral dissertation). (In Korean).
- Mun, Suk Yeong. 2009. *Hankwukeuy sicey pemcwu* [The Korean tense category]. Seoul: Thayhaksa. (In Korean).
- Na, Chin-Seok. 1953. Milaysisangpocoekan *li-wa keyss-uy kyocheu* [the changes of future tense-aspect auxiliary stems *li* and *keyss*]. *Kwuke Kwukmwunhak* 6. no pages. Reprinted in Na, Chin-Seok, 1971. *Wulimaluy ttaymaykim yenkwu* [A study on the Korean tense system], 296–306. Seoul: Kwahaksa. (In Korean).
- Nam, Ki Shim. 1972. Hyuntaykwuke siceyey kwanhan yenkwu [a study of the tense system in modern Korean]. *Kwuke Kwukmwunhak* 55-57. 213–238. Reprinted in Nam, Ki Shim, 1978, *Kwukemwunpepuy sicey mwunceyey kwanhan yenkwu* [A study of the tense system in Korean grammar], 1–34. Seoul: Tower Press.
- Narrog, Heiko & Seongha Rhee. 2013. Grammaticalization of space in Korean and Japanese. In Martine Robbeets & Hubert Cuyckens (eds.), *Shared grammaticalization with special focus on the transeurasian languages*, 287–315. Amsterdam: John Benjamins.
- Narrog, Heiko, Seongha Rhee & John Whitman. 2018. Grammaticalization in Japanese and Korean. In Heiko Narrog & Bernd Heine (eds.), *Grammaticalization from a typological perspective*, 166–188. Oxford: Oxford University Press.
- Park, Jinho. 1998. Kotay kwuke mwunpep [a grammar of old Korean]. In The National Institute of the Korean Language (ed.), *Kwukeyu sitaypyel pyenchen*

- yenkwu 3 [A study of historical change of Korean 3], 121–205. Seoul: The National Institute of the Korean Language.
- Park, Seung-Bin. 1972[1935]. *Cosenehak* [Korean linguistics]. Seoul: Tongmunkwan Publishing.
- Partee, Barbara Hall. 1973. Some structural analogies between tenses and pronouns in English. *The Journal of Philosophy* 70. 601–609.
- Ramm, Wiebke. 2008. Upgrading of non-restrictive relative clauses in translation: A change in discourse structure? In Cathrine Fabricius-Hansen & Wiebke Ramm (eds.), *'subordination' versus 'coordination' in sentence and text: A cross-linguistic perspective*, 135–159. Amsterdam: John Benjamins.
- Ramstedt, Gustaf J. 1903. *Über die konjugation des khalkha-mongolischen*. Helsingfors: Société Finno-Ougrienne.
- Ramstedt, Gustav John. 1997[1939]. *A Korean grammar*. Helsinki: Suomalais-Ugrilainen Seura.
- Rhee, Seongha. 1996. *Semantics of verbs and grammaticalization: The development in Korean from a cross-linguistic perspective*. PhD dissertation. The University of Texas at Austin. (Doctoral dissertation).
- Rhee, Seongha. 2008. On the rise and fall of Korean nominalizers. In Maria José López-Couso & Elena Seoane (eds.), *Rethinking grammaticalization: New perspectives*, 239–264. Amsterdam: John Benjamins.
- Rhee, Seongha. 2011. Nominalization and stance marking in Korean. In Foong Ha Yap & Janick Wrona (eds.), *Nominalization in Asian languages: Diachronic and typological perspectives*, 393–422. Amsterdam: John Benjamins.
- Rhee, Seongha. 2012a. Context-induced reinterpretation and (inter)subjectification: The case of grammaticalization of sentence-final particles. *Language Sciences* 34. 284–300. DOI: 10.1016/j.langsci.2011.10.004.
- Rhee, Seongha. 2012b. Dimensions and force dynamics in perception and grammar: A grammaticalization perspective. *Studies in Modern Grammar* 70. 181–206.
- Rhee, Seongha. 2020. Grammaticalization in Korean. In Walter Bisang & Andrej Malchukov (eds.), *Grammaticalization scenarios: Cross-linguistic variation and universal tendencies: Grammaticalization scenarios from Europe and Asia*, vol. 1, 575–608. Berlin: de Gruyter.
- Rhee, Seongha & Tania Kuteva. Forthcoming. Apprehensional markers in Korean. In Marine Vuillermet Martina Faller & Eva Schultze-Berndt (eds.), *Apprehensional constructions in a cross-linguistic perspective*. Berlin: Language Science Press.

- Robbeets, Martine, Remco Bouckaert, Matthew Conte, Alexander Savelyev, Tao Li, Deog-Im An, Ken-ichi Shinoda, Yinqiu Cui, Takamune Kawashima, Geonyoung Kim, et al. 2021. Triangulation supports agricultural spread of the transeurasian languages. *Nature* 599. 616–621. DOI: 10.1038/s41586-021-04108-8.
- Robbeets, Martine, Mark Hudson, Chao Ning, Remco Bouckaert, Alexander Savelyev, Geonyoung Kim, Tao Li, Sofia Oskolskaya, Ilya Gruntov, Olga Mazo, et al. 2022. Triangulation reduces the polygon of error for the history of transeurasian. *bioRxiv preprint*. DOI: 10.1101/2022.10.05.510045.
- Ryu, Yeol. 1992. *Cosenmal lyeksa* [A history of the Korean language], vol. 2. Pyeongyang: Social Science Academy Press. (In Korean).
- Shin, Chang Soon. 1972. Hyentay hankwuke yongen pocoekan keyss-uy uyuywa yongpep [on the semantics and uses of auxiliary -keyss- in modern Korean]. *Cosenhakpo* 65. 119–140. (In Korean).
- Sohn, Ho-Min. 1975. Retrospection in Korean. *Language Research* 11(1). 87–103.
- Sohn, Ho-Min. 1999. *The Korean language*. Cambridge: Cambridge University Press.
- Song, Jae Jung. 2005. *The Korean language: Structure, use and context*. London: Routledge.
- Suh, Cheong Soo. 1976. Kwuke sisanghyengthayuy uymi pwunsek yenkwu. *Mwunpepyenkwu* 3. 83–158. (In Korean).
- Suh, Cheong Soo. 1977. Keyss-ey kwanhayae [On keyss]. *Mal* 2. 63–87.
- Sung, Ky-Chul. 1976. -Keyss-kwa -ul kesi-uy yumi pikyo [a comparison of the semantics of -keyss- and -ul kesi-]. In Publication committee (ed.), *Kimhyengkyu kyoswu cengnyenthoyim kinyem nonmwun* [Festschrift for retirement of Professor Kim Hyung Kyu], 223–232. Seoul: Seoul National University Press. (In Korean).
- Tian, Zheng, Yuxin Tao, Kongyang Zhu, Guillaume Jacques, Robin J Ryder, José Andrés Alonso de la Fuente, Anton Antonov, Ziyang Xia, Yuxuan Zhang, Xiaoyan Ji, et al. 2022. Triangulation fails when neither linguistic, genetic, nor archaeological data support the transeurasian narrative. *bioRxiv*. DOI: 10.1101/2022.06.09.495471.
- Tsunoda, Tasaku (ed.). 2013. *Adnominal clauses and the ‘mermaid construction’: Grammaticalization of nouns*. Tokyo: NINJAL.
- Underwood, Horace Grant. 1890. *An introduction to the Korean spoken language*. Yokohama: Kelly & Walsh.
- Yae, Sunhee, Tania Kuteva & Seongha Rhee. 2023. From fear to grammar: The case of avoidives. *Lingua* 288. 1–22.

- Yap, Foong Ha, Pik-ling Choi & Kam-siu Cheung. 2010. Delexicalizing *di*: How a Chinese noun has evolved into an attitudinal nominalizer. In An Van linden, Jean-Christophe Verstraete & Kristin Davidse (eds.), *Formal evidence in grammaticalization research* [Typological Studies in Language 94], 63–91. Amsterdam: John Benjamins.
- Yap, Foong Ha & Jiao Wang. 2011. From light noun to nominalizer and more: The grammaticalization of *zhe* and *suo* in old and Middle Chinese. In Foong Ha Yap & Janick Wrona (eds.), *Nominalization in Asian languages: Diachronic and typological perspectives*, 61–107. Amsterdam: John Benjamins.
- Yeom, Kwang Ho. 1998. *Congkyelemiuy thongsicek yenkwu* [a diachronic study of sentential endings]. Seoul: Pakiceng Publishing.
- Yoo, Chang-Shik. 1956. Siceycheykyeyey kwanhan silon [thoughts on the tense system]. *Journal of Kyungpook National University* 1. 91–159. (In Korean).